

Satyabrata Das

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Summary

Computer Vision and ML Engineer with 3+ years shipping production software, now building real-time perception systems — eye tracking, gaze estimation, 3D pose — at 40 FPS on consumer hardware. M.S. Artificial Intelligence Systems, University of Florida, May 2027.

Education

University of Florida

M.S., Artificial Intelligence Systems

Aug 2025 – May 2027

Gainesville, FL

- Coursework: Machine Learning for AI, Applied Deep Learning, AI Systems, AI Safety & Robustness, Computer Vision.

Experience

Evyy — Computer Vision Engineer, Eye Tracking & ML

May 2026 – Present

- Shipped real-time eye tracking, iris detection, and blink detection for AI interview monitoring using Python, OpenCV, and MediaPipe, sustaining 40 FPS real-time inference on standard consumer webcams.
- Improved gaze estimation accuracy and model robustness under glasses glare, off-axis head pose, and low-light conditions through calibration validation and adaptive thresholding, validated across 50 test configurations.
- Developed deep learning anomaly detection models in PyTorch and TensorFlow to classify behavioral patterns from three signal streams: gaze direction, iris movement, and blink cadence.
- Optimized browser-based model inference and deployment pipelines to hold real-time frame rate on low-end hardware; owned the first production release end to end with the founding team.

University of Florida — Machine Learning Research Assistant

Mar 2026 – May 2026

- Constructed video-based kinematic analysis pipelines for Parkinson's assessment using MediaPipe 3D pose estimation across 1,000+ clinical movement recordings.
- Trained regression models mapping biomechanical features (joint angles, angular velocity, left-right symmetry) to clinician-rated severity scores, replacing manual video review.

SouthEnd Psychiatry — IT & Operations Contractor

Sep 2024 – Aug 2025

- Administered EHR and practice-management systems for a telehealth psychiatry group of 20 clinicians and 10,000+ patient records under strict HIPAA requirements.
- Automated intake reconciliation in Python, eliminating 30 hours of manual entry weekly and cutting scheduling conflicts 25% across 5,000+ monthly appointments.

ARC Document Solutions — Software Engineer

Jul 2022 – Sep 2024

- Engineered real-time WebSocket sync layer for ARC Facilities (20K+ MAU), cutting collaboration latency 25% and support tickets ~30% quarter-over-quarter.
- Integrated 5 payment gateways (Razorpay, Paytm, Cashfree, Braintree, Elavon) and reCAPTCHA into ARC Print, reducing failed transactions 15%.
- Rebuilt the data and concurrency layer with Core Data and GCD multithreading, cutting app load time 30% and eliminating UI freezes flagged in QA.

SCI-BI Software Pvt Ltd — Software Trainee

Jul 2021 – Dec 2021

- Delivered Python/SSIS ETL pipelines handling 10M+ records daily at 99%+ accuracy, cutting 20 manual hours weekly.
- Tuned SQL queries and transformations with Prometheus/Grafana monitoring, reducing report generation time 40% and pipeline failures 25%.

Projects

SummarIQ — NLP Pipeline for Document Intelligence

GitHub

- Designed NLP pipeline using HuggingFace Transformers (T5/BART) and LangChain for classification and semantic retrieval over 200+ documents, improving accuracy ~18%.
- Deployed via FastAPI at sub-1.4s latency with FAISS chunking and embedding strategies for scalable semantic search.

JuggleIQ — CV Soccer Juggling Coach

GitHub

- Architected a YOLO + MediaPipe + Kalman-filter pipeline that detects the ball, tracks pose, counts left/right foot touches, and returns skill metrics from video.
- Served it from a GPU-backed FastAPI service with a React + Vite + Recharts dashboard showing skill score, rhythm, and touch timeline with annotated replay.

Technical Skills

- **AI & ML:** PyTorch, TensorFlow, HuggingFace Transformers, Scikit-learn, LLMs, NLP, RAG, CUDA, FAISS, vector DBs.
- **Computer Vision:** OpenCV, MediaPipe, YOLO, 3D Pose Estimation, Eye Tracking, Gaze Estimation, Kalman Filtering.
- **Tools:** Python, C++, Swift, SQL, TypeScript, FastAPI, Docker, REST APIs, CI/CD, Git, Linux.